SPECIFICATION SHEET

FOR

SCISSOR LIFT

RFQ - ENQ.100160 - R0

PROJECT NO. ENQ.100160

REVISIONS

Rev	Description	Date	Author
00	First Issue	06.02.2020	GCLA

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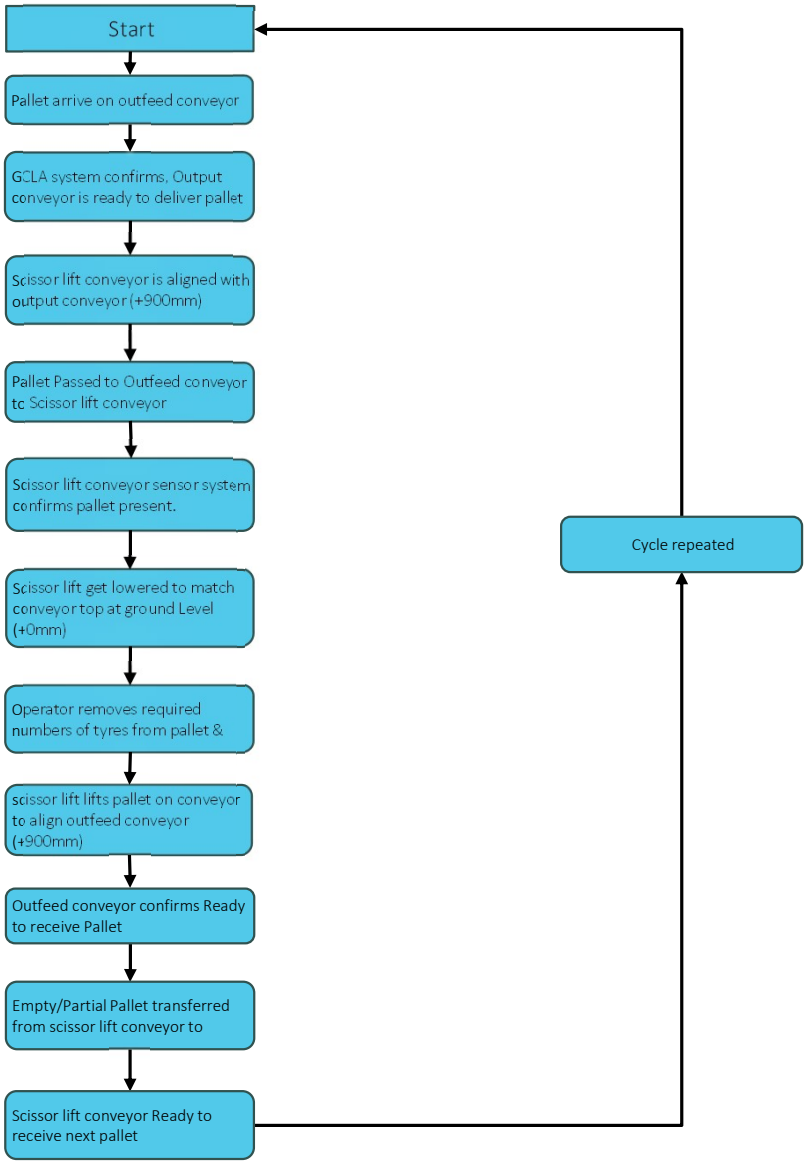
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1.0 Description:

Scissor Lift is automatic Lifting and lowering device, handles Pallets at outbond area

1 Pit Mounted Scissor Lift
2 QTY: 03 no.

2.0 Sequence of Operation

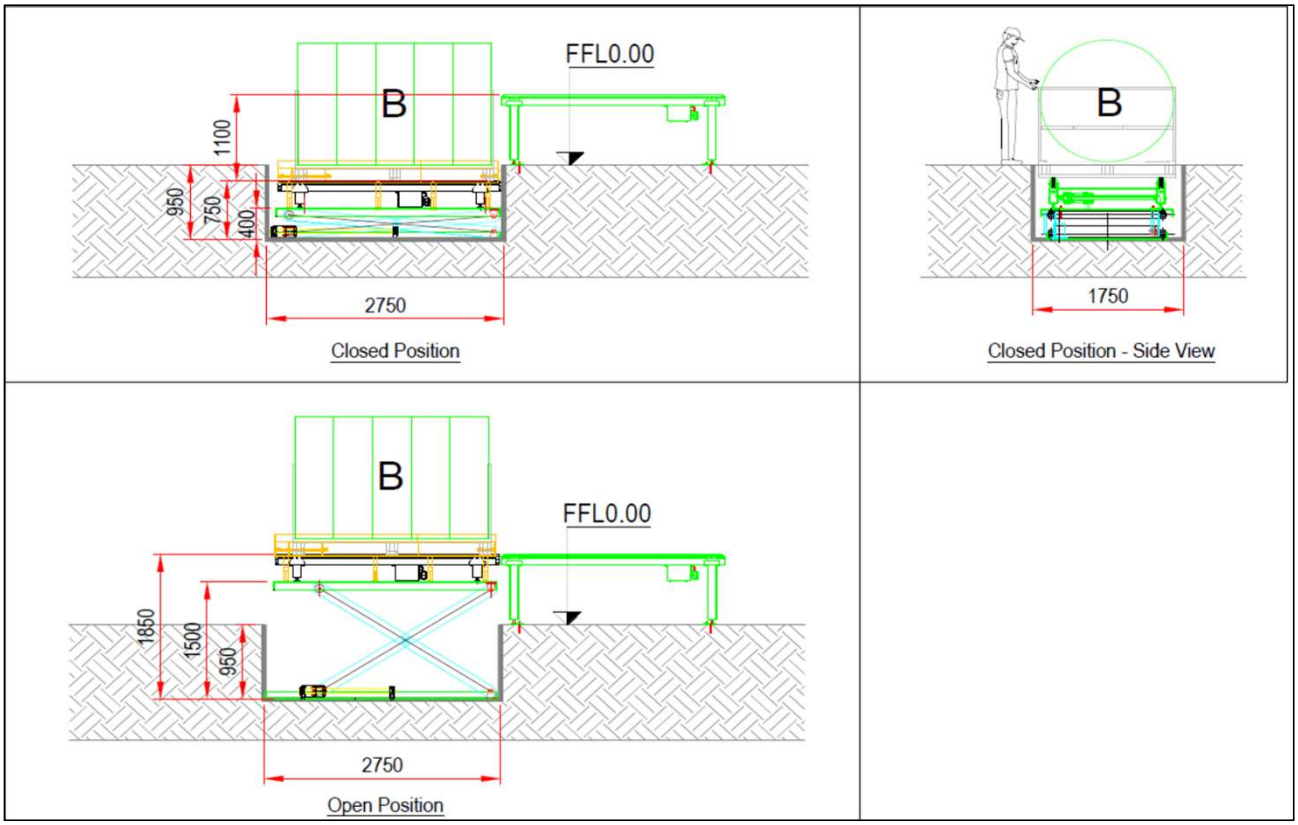


3.0 Technical specification:

1	Unit Handled	Pallets on chain conveyor
2	Pallet Dimensions	2300mm(W) x 1600mm(D) x 1800mm(H)
3	Weight of Pallet (Max)	1200 Kg (Including pallet self wt.)
4	Conveyor Dimensions	1350mm(W) x 2600mm(L) x 350mm(H)
5	Weight of Conveyor (Max)	300 Kg
6	Scissor Lift Cycle time	25 Cycles per hour
7	Speed of the conveyor (scope GCLA)	0.3 m/s
8	Acceleration of the conveyor (scope GCLA)	0.3 m/s2
9	No of Positions	2 nos (Closed Position - 400 mm and Open Position - 1500 mm)
9	Positioning Accuracy	+/- 5mm
10	Load to Lift	1400 Kg
11	Load to Lower	1400 Kg
12	Top Platform Size	2700 mm(L) x 1500 mm(W)
13	Maximum Height	+ 1490 mm
14	Minimum Height	+ 400 mm
15	Lifting Stroke	1100 mm
16	Load Distribution	Uniform
17	Control type	PLC + Push Button
18	Scissor Lift Mobility	Fixed Type
19	Power pack Position	Inside
20	Top Platform	MS structure with Chequrred Plate Platform
21	Lifting speed	100 to 150 mm/sec
22	Lowering speed	100 to 150 mm/sec
23	Level Accuracy	+/- 2 mm
24	Lowering speed	100 to 150 mm/sec
25	Cycle time	25 JPH for 4 sets
26	Space available	refer Drawing No. 01
27	Floor requirement	Floor with suitable strength for static and dynamic loads of palletiser with sufficinet depth for grouting will be provided by end customer. Supplier to provide complete details as mentioned in the drawing.
28	No of Shifts	3 Shift operation (1 Shift = 8Hrs)
29	Availability of Scissor Lift	> 98%

4.0 G A Layout

Drawing No - 01 Scissor Lift working drawing - For Reference Only



5.0 Scope of supply - Supplier

1 Scissor Lift as per "3.0 Technical Specifications"
2 Supervision for installation and signed off.
3 Structural support members, if any
4 Interface with Safety fencing,
5 Interface with Siemens PLC (GCLA Scope) for error pallletizer display, Consider Potential free contact - 02 Nos.
6 Integration with sensors for detection of pallet
9 Operator Panel with PB box
10 All required safety feature and accessories as per industry standard
11 All anchoring fasteners for machine (Chemical Anchoring)
12 Cement for levelling and grouting, If required.
13 Mounting arrangement for Cable drag chain. (If any)
14 All sensors & limit switches including safety switches / devices
15 All Cables required to terminate sensors / devices in GCLA JB
16 Multipin connectors at JB for connecting sensors & devices
17 Cable routing & trays with cover wherever required
18 All sensors & devices termination to be done at supplier factory & to be tested jointly
19 All sensors & devices should be connector type, screw connection / soldering is not allowed.
22 All required Hydraulic components
23 Pressure and flow measurement and feedback system
25 Calibration and Lubrication requirement and details/Chart
26 Commissioning spares, Consumables & wear parts
27 Integrated Performance demo, PAC & FAC.
28 Whatever is not included in GCLA scope.

6.0 Safety & interlock requirement

1 Interlock with GCLA Conveyor system
2 E stop Button with illuminated base
3 Jerk free movement
4 Toe trap prevention device/ Toe guard
5 Bellows for lower part of platform
6 Floor Barricades

7.0 GCLA Scope

1 Outfeed pallet handling conveyor
2 Pallet handling Conveyor on scissor Lift
3 Sensors for pallet start stop
4 Pallet present detection.
6 PLC Programming for GCLA conveyors
8 Unloading of material from truck to storage point, which is 20 mtr away from installation point.

Note -

1 Semi - skilled Labour for installation - Will be confirmed after discussion between GCLA & Supplier
2 Material handling equipment for installation - Will be confirmed after discussion between GCLA & Supplier

8.0 Equipment Finish requirements

Fabrication

1 All spatters to be removed.
2 Welding should be uniform.
3 No blow holes in welding.
4 All critical welds as identified by GCLA shall be checked by DP test.

Machining

1 All machined surfaces shall be protected with corrosion protective coating.
2 Remove all sharp edges.

Painting

Process	Spray painting
Paint base	Epoxy - One coat of primer +2 coats of finished paints
Cleaning process	Sand blasting as per IS 1477
Finishing	Glossy finish
Paint thickness	120 - 150 microns

Powder coating

Process	Powder coating
Paint base	Epoxy
Cleaning process	7 tank process
Finishing	Glossy finish
Paint thickness	60 - 80 microns

Plating

Process	Trivalent chromate plating
Thickness	10 - 20 microns

Colours

1 Stationary structures: RAL 5003
2 Moving parts: RAL 1028

Electrical Wiring

1. All wiring should in compliance with IEC standards
2. Multicore cables with Wires numbering pattern to be used

IS Standards to be Followed

IS 800 – 2007	General Construction on steel
IS 808 – 1989	Dimensions for hot rolled steel Beam, Column, Channels & angle sections
IS 3655 – 1966	Recommended practice for electroplating
IS 10748 – 1984	Hot-rolled Steel Strip for Welded Tubes and Pipes - Specification
IS 101 – 1964	Methods of Sampling and test for paints, varnishes and related products
IS 2074 – 1992	Ready Mixed Paint, Air Drying, Red Oxide Zinc Chrome, Priming - Specification
IS 513	Cold Rolled Low Carbon Steel Sheets & Strips
IS 325 – 3	Phase induction motor-Specifications.
IS 1363	Hexagon Head Bolts, Screws and Nuts
IS 2062 – 1999	Steel for General Structural Purposes – Specification
IS 4923 – 1997	Hollow Steel Sections for structural use.
IS 2016	Specification for plain washer.

9.0 Site Address

Aurangabad, Maharashtra, India

10.0 Site Location & Climatological data

	Item	Value	
1	Ambient temperature ^{0}C	Max	
		Min	
2	Design temperature ^{0}C	Min 45	
3	Relative Humidity	Max	
		Min	
4	Wet bulb temperature	Max	
		Min	
5	Environment		
6	Rain fall, Avg annual mm		
7	Wind, max velocity m/s		
8	Earthquake	As per IS	
9	Plant side elevation above MSL		
10	Topography at site	Flat	
11	Drainage pattern		
12	Nearest highway		
13	Nearest village/town/city		
14	Seismicity		

11.0 Tentative Delivery schedule

		Date
1	PO from GCLA	To be Discussed
2	Drawing approval	To be Discussed
3	FAT	To be Discussed
4	Dispatch	To be Discussed
5	Delivery at Site	To be Discussed
6	Installation & Commissioning	To be Discussed
7	Integrated Performance Demo	To be Discussed
8	Training and handover	To be Discussed

12.0 List of Approved makes

	Description	Make
1	Geared Motors & VFD	Sew Euro drive
2	Hydraulic Power pack	Euton/OEM make/Equivalent
3	Hydraulic Cylinders	Euton/OEM make/Equivalent
4	PLC, HMI & Switchgear	Siemens
5	Push Buttons	Siemens /Equivalent
6	Limit Switches	Omron
7	Sensors	Sick / Leuze / P&F
8	Bearings*	SKF / INA / FAG
9	Chains	Ti Diamond / Rollon
10	Power & Communication Cables	Igus / Laap
11	Anchor fasteners for grouting	HILTI
12	Fasteners	TVS / Unbrako
13	Structural steel	TATA/JINDAL/SAIL/ESSAR/ Any international make

Note: All bought out components should be of reputed make. Supplier should take prior approval from GCLA for make of items not covered in above list.
All critical fasteners should be of 12.9 grade. Other fasteners should be minimum 8.8 grade.
* All bearing shall be both side sealed bearing 2RS / 2Z & lifespan should be minimum 5 years with 99% reliability

13.0 DAP Requirement

Sr. No.	Description	Date	Approver from GCLA
1	GA Drawing in AutoCAD & PDF.		Mechanical Design
2	Floor load with static, dynamic and uplifting forces /moments in closed and extended condition		Mechanical Design
3	Utility, power & air requirement		Mechanical Design, Automation Design
4	Power requirement		Automation Design
5	compliance with GCLA RFQ		Mechanical Design, Automation Design
6	Approval on RFQ deviataion if any		Mechanical Design, Automation Design
7	Civil requirement like Grouting details and finish floor level		Mechanical Design
8	Spares parts list with separate price with critical & desirable distribution		Mechanical Design, Automation Design
9	List of consumable items with separate price, if any		Mechanical Design, Automation Design
10	Input/Output/Interface details		Automation Design
11	Electric BOM Including Sensors & devices BOM as per GCLA format		Automation Design
12	Complete Datasheets for all geared motor & drives		Mechanical Design, Automation Design
13	Complete Datasheets for all pneumatic components		Mechanical Design, Automation Design
14	Complete Datasheets for all sensors & devices		Automation Design
15	Project Specific Installation Guidelines document		Mechanical Design, Automation Design, Software Design, Installation & Quality Control
16	Integrated Performance Demo procedure & Acceptance criteria		Mechanical Design, Automation Design
17	Quality Assurance plan. Approval on QAP from GCLA quality team is mandatory before Finalization.		Mechanical Design, Automation Design, Quality Control
18	Test certificates of all boughtout items		Mechanical Design, Automation Design Quality Control

**Note - Dates for DAP submission will be confirmed after discussion between GCLA & Supplier

14.0 Acceptance criteria & performance criteria

	Machine Qualification	Document/criteria
1	Performance test	Approved drawing
2	Cycle speed = 144 sec	For min 50 Cycles
2	leveling accuracy	+/- 2 mm
3	Overall size	Approved drawing
4	Specification & makes validation of bought outs	Approved make list
5	Surface finish, aesthetic	As per specification sheet
6	Completeness of assembly	Approved drawing
7	Special requirement if any, as per approval drawing	Approved drawing
8	Scope of supply as per specification sheet	As per specification sheet
9	Closure of all electrical & mechanical safety hazards	As per applicable norms
10	Noise level	As per applicable norms
11	Power consumption	As per specification approved drawing
12	Functional Parameters of motors and drives	As per specification / approved drawing
13	Functional Parameters of Hydraulic systems	As per specification / approved drawing
14	All error demonstration & training	Test to be performed & Certificate to be issued
15	Test certificates for all bought out items like Motor , chains, sprocket, shaft, Power pack etc.	Test Certificates
16	Integrated Performance at PAC level	90% of Defined cycle for 4 hours
17	Integrated Performance at FAC level	100% of Defined cycle for 4 hours

	Mechanical functional check list	permissible values
1	Lifting and lowering test	Operation test , Visual
2	Structural stability	Operation test , Visual
3	Maintenance access to all bearing, motor, limit switches.	Operation test , Visual
4	Emergency access to motor & junction Box.	Operation test , Visual
5	Lubrication point availability and its access level.	Operation test , Visual
6	Effect of overshoot in case of electrical failure.	Operation test , Visual
7	Functionality of all Safety features	Operation test , Visual
8	System uptime	Above 98%

	Electrical functional check list	permissible values
1	Current drawn for Electrical motor without load, With working load and with Max load.	As per motor specification
2	IO check	As per IO list & BOM
3	VFD integration and compatibility to system.	Operation test , Visual
4	Wiring & Cable routing check	IEC standards, Visual
5	JB mounting arrangement check	Visual
6	Multipin connector mounting & termination check	Visual & As per electrical drawings

15.0 Terms & Condition

1. FAT : All equipment must be offered for FAT to check performance parameters. In case it is not possible to check all parameters in FAT it should be
2. Confidential Information: Supplier agree that any information received pursuant to this inquiry shall be treated as confidential information and shall not be disclosed to any third party except as is required by law or legal process.
3. List of Critical Spare Parts: Supplier should provide list of spare parts along with quote which are critical for trouble free functioning of the equipment for minimum period of three years after warrantee period.
4. Spare Parts Availability: Supplier should guarantee the availability of spare parts for the equipment for period of 15years from date of supply.
5. Supplier shall provide training to GCLA engineers on adjustment, maintenance and operation of all components.
6. Material Unloading in the scope of supplier.
7. Labors, technicians & supervision in the scope of supplier.
8. Statutory compliance requirements as per law of land and labour laws in scope of supplier.
9. Site Safety rules & regulations should be followed by supplier.
10. Equipment hiring(like Hydra, Mobile Crane, Forklift) in scope of supplier.
11. Cement padding after levelling of machine in the scope of supplier, If Applicable

Note: Suppliers has to agreed all the points mentioned in RFQ. If there is any deviation, supplier need to take approval from GCLA before sending the offer

16.0 Abbreviations

GCLA: Godrej Consoveyo logistics Automation Ltd
PLC: Programmable Logic Controller
QAP: Quality Assurance Procedure
BOM: Bill of material
MOC: Material of Construction
PAC: Preliminary Acceptance Certificate
FAC: Final Acceptance Certificate

17.0 RFQ Approvals

Mechanical Design	Automation Design	Software Design
Procurement	Quality Department	Installation
Project Head	Engineering Head	Sales Head